Phishing Emails Analysis Report

Introduction

Phishing is a type of cyber attack where attackers attempt to trick users into revealing sensitive information such as passwords, banking details, or OTPs. These attacks are commonly carried out through emails that appear to be from legitimate organizations.
Phishing attacks rely heavily on social engineering techniques such as urgency, fear, and trust to manipulate users into taking immediate action. Due to the increasing number of phishing attacks, it is important for users and organizations to understand how to identify and prevent them.

Objective

The objective of this project is to analyze phishing email samples, identify common phishing indicators, classify the risk level of emails, and create awareness guidelines to help users avoid phishing attacks.

Tools Used

- Google Admin Toolbox Messageheader – for analyzing email headers
- MXToolbox – for verifying sender and domain information
- Web Browser – for inspecting URLs and identifying suspicious links

Methodology

The analysis was performed by collecting phishing email samples and examining their subject lines, content, sender details, and links.
Header analysis was also conducted using email analysis tools to identify authentication failures such as SPF, DKIM, and DMARC.
Based on these observations, phishing indicators were identified and the risk level of each email was classified.

Email Analysis

Email 1

Subject: Urgent: Your Account Will Be Locked

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

Verify Now: http://secure-account-verify.com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

Analysis

- The sender appears suspicious and does not belong to an official domain. Attackers often use fake

email addresses to trick users.

- The subject uses urgent language like “Urgent” and “Account Locked” to create panic and force immediate action.
- The email uses a generic greeting (“Dear User”) instead of the recipient’s name, which is a common phishing technique.
- The link provided is not from an official domain and may redirect users to a fake website to steal credentials.

Phishing Indicators:

1. Fake/suspicious sender
2. Urgent language
3. Generic greeting
4. Suspicious link

Risk Level:

Phishing (High Risk)

Email 2

Subject: New Fax Message

Service: SharePoint

You have received a new fax message.
Click the link below to view the document.

Print & Preview Here: https://attacker.example.com/

Analysis:

- The sender appears suspicious and does not belong to an official SharePoint or fax service provider.
- The subject is vague and does not clearly indicate the purpose, which can confuse users.
- The email claims that a fax has been received and asks the user to click a link, which is a common phishing technique.
- The link redirects to an unknown domain (attacker.example.com), which is not a trusted or official source.

Phishing Indicators:

1. Unexpected fax notification
2. Suspicious line
3. Unknown sender
4. Social engineering

Risk Level:

Phishing (High Risk)

Email 3

Subject: New Voice Message Notification

Service: Voicemail System

Internal Notification: Voicemail Received

You have received a new voice message.

Click here to listen: http://attacker.example.com

Analysis:

- The sender appears to imitate a legitimate voicemail service but does not belong to a trusted domain.
- The subject mentions a voice message, which creates curiosity and encourages the user to open the email.
- The email claims that a voicemail has been received and prompts the user to click a link, which is a common phishing technique.
- The link leads to a suspicious domain (attacker.example.com), which may be used to steal user data or install malware.

Phishing Indicators:

1. Fake notification
2. Social engineering (curiosity-based attack)
3. Suspicious link
4. Unknown sender

Risk Level:

Phishing (High Risk)

Email	**Risk Level**
Email 1	Phishing (High Risk)
Email 2	Phishing (High Risk)
Email 3	Phishing (High Risk)

Conclusion

Phishing attacks are one of the most common cybersecurity threats and can lead to serious data breaches. This project demonstrated how phishing emails can be identified using both content analysis and header analysis techniques.

Awareness and cautious behavior are essential to prevent such attacks and ensure cybersecurity.
This project highlights the importance of both technical and user-level awareness in defending against phishing attacks.